

## **CHE 455 – Nanoscale Systems in Chemical Engineering**

Instructor: Naveen Dandu

This assessment reflects on the presentations delivered by students in **CHE 455: Nanoscale Systems in Chemical Engineering**, where they explored Scanning Tunneling Microscopy (STM) as a tool for understanding matter at the atomic scale. The goal was to evaluate their technical understanding and their ability to communicate complex scientific ideas.

Students were asked to present on STM, covering its physical principles, instrumentation, and applications. In particular, they were expected to explain quantum tunneling, describe how STM achieves atomic resolution, and connect theory with real-world applications such as surface imaging and electronic structure analysis.

What stood out most was the clarity with which students explained STM concepts. They made good use of diagrams, maintained a logical flow, and demonstrated confidence in presenting technical material. Importantly, they were able to connect abstract ideas, like quantum tunneling, to tangible outcomes such as atomic-scale imaging.

While the overall performance was strong, there is room to deepen the level of understanding. Future presentations could benefit from more emphasis on the underlying physics for instance, distinguishing between topographic and electronic contributions to STM images, or discussing the role of local density of states (LDOS) in greater detail. There is also an opportunity to encourage more critical thinking by asking students to analyze limitations or interpret experimental scenarios.

### **Rubric Assessment Summary:**

On an average, students received a score of 2.81 (Overall Satisfactory), with a variation of 0.31. While some indicators show variation, others are perfectly uniform. This suggests that students have a shared baseline competency, likely supported by course structure and guidance. At the same time, the variation in a few indicators highlights where individual differences in skill development still exist.